

Secure Browser-to-Browser File Sharing Platform using WebRTC and Socket.io

Abstract

This project implements a secure browser-to-browser file sharing system using WebRTC. The application enables direct peer-to-peer transfer of files while ensuring confidentiality and integrity through AES-GCM encryption and SHA-256 hashing. Socket.io is used as a signaling mechanism to establish WebRTC connections between peers.

Objective

To develop a secure and efficient file sharing platform that allows users to transfer files directly between browsers without relying on centralized storage.

Technologies Used

Frontend:

- React
- Vite
- Tailwind CSS

Backend:

- Node.js
- Express.js
- Socket.io

Communication:

- WebRTC
- Simple-Peer

Security:

- Web Crypto API
 - AES-GCM Encryption
 - SHA-256 Hashing
-

System Architecture

Sender Browser ↓ Socket.io Signaling Server ↓ Receiver Browser

Peer-to-Peer WebRTC Channel

Features Implemented

1. Browser-to-Browser File Transfer
 2. Socket.io Signaling Server
 3. AES-GCM End-to-End Encryption
 4. SHA-256 Integrity Verification
 5. Chunk-Based File Transfer
 6. Progress Monitoring
 7. Transfer Speed Calculation
 8. ETA Prediction
 9. Resume Support
 10. QR Code Sharing
 11. Error Handling
 12. Transfer History
 13. Dark Mode User Interface
-

Working Procedure

1. Sender selects a file.
2. SHA-256 hash is generated.
3. AES-GCM encryption key is created.
4. Room is created through Socket.io.
5. Receiver joins using room ID.

6. WebRTC establishes a direct connection.
 7. File chunks are encrypted and transmitted.
 8. Receiver decrypts chunks and reconstructs the file.
 9. SHA-256 verification is performed.
 10. File is downloaded automatically.
-

Results

The application successfully performs peer-to-peer file transfer with encryption and integrity verification. Progress tracking and transfer statistics improve user experience while reconnect mechanisms increase reliability.

Applications

- Secure document sharing
 - Educational file exchange
 - Corporate collaboration
 - Privacy-preserving communication
-

Future Scope

- TURN server integration
 - Multiple file transfer
 - Folder sharing
 - File compression
 - Mobile support
-

GitHub Repository

<https://github.com/aditya7600/p2p-web-share>

Conclusion

The Secure Browser-to-Browser File Sharing Platform demonstrates the use of WebRTC and Socket.io for efficient peer-to-peer communication. The project successfully combines encryption, integrity verification, and modern web technologies to provide a secure and user-friendly file sharing solution.